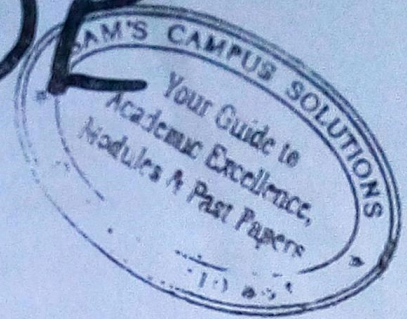


EEE 102



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2014/2015

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN ELECTRICAL AND ELECTRONICS ENGINEERING

EEE 102: COMPUTER PROGRAMMING I

DATE: Tuesday, 14th April 2015

TIME: 2.00 p.m. – 4.00 p.m.

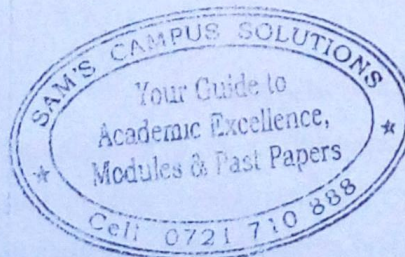
INSTRUCTIONS:

- Attempt Question ONE and any other TWO Questions

Question One [30 Marks]

- ✓ a) Using appropriate examples distinguish between machine language, assembly language and high-level languages. [3 Marks]
- ⊗ b) Write a C function named MyPower that receives two values v and p , and returns the value of v^p . Note: implement your function using recursion. [3 Marks]
- ✓ c) What is an identifier as used in C programming? What rules must be observed when creating an identifier? [3 Marks]
- ✓ d) Develop a flow chart for the following piece code. [3 Marks]

```
int list[6];
for(int j = 0; j < 6; j++)
{
    list[j] = 3 * j + 4;
    if (j % 3 == 0)
    {
        list[j] = list[j] - 2;
    }
}
```



- ✓ e) "C is a compiled language." Explain this statement giving a detailed description of a compiler and the compilation and linking process. [3 Marks]

- ✓ f) Write an algorithm for a C program that accepts student marks as inputs and assigns them grades.

The grades are assigned as follows:

Above or equals 70 marks:	A
Between 60 and 69 marks:	B
Between 50 and 59 marks:	C
Between 40 and 49 marks:	D
Less than 40:	E

The program should not allow an input of more than 100 or less than 0 marks. [4 Marks]

- ✓ g) A declaration tells the compiler the name and type of variable you'll be using in your program. Provide THREE syntactically correct formats that can be used to declare a variable. Explain the various fields in the format. [3 Marks]

- ⊗ h) What are the advantages of using modular approach in program design? [3 Marks]

- ✓ i) What is the output of the following code segment? [2 Marks]

```
sum = 0;
for(i = 1; i < 5; i++)
    for(j = 10; j < 15; j++)
        for(k = 0; k < 5; k++)
            sum += i + j + k;
printf("%d\n", sum);
```

1, 2, 3, 4
10 11, 12, 13, 14
0 1 2 3 4

- ⊗ j) Provide a detailed comparison of structured programming languages and object-oriented programming languages. [3 Marks]

Question Two [20 Marks]

- ✓ a) Write the pseudo-code for an ATM machine that strictly allows only the following transactions to be carried out by the user: check account balance, withdraw money, buy airtime and deposit cash.
• Withdrawals and airtime purchases require that there must be sufficient balance in the account. The system should not allow any ATM transaction of more than KES 20,000. All transaction require PIN authentication and any three consecutive wrong PIN inputs results in retention of the users ATM card. [6 Marks]



EEE 102

KENYATTA UNIVERSITY

School of Engineering

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EEE 102: COMPUTER PROGRAMMING I

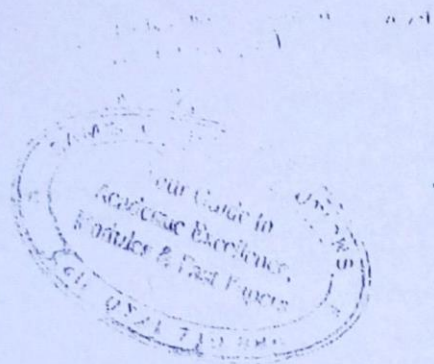
CAT 1 [70 Marks]

DATE: 18th February, 2015

TIME: 0900 - 1030

- ✓ 1. What is an identifier as used in C programming? What rules must be observed when creating an identifier? [3 Marks]
- ✓ 2. Discuss the generalized methodology involved in the problem solving. [8 Marks]
- ✓ 3. What is a variable? What are the rules for defining variables? Differentiate between local variable and global variable. [3 Marks]
- ✓ 4. Discuss the primary data types in C. [4 Marks]
- ✓ 5. Write the pseudo-code, algorithm, flowchart and an introductory program code for a C Program to print all numbers between 1 to n that are divisible by 7. [12 Marks]
- ✓ 6. Write the pseudo-code and an algorithm that reads three numbers, and prints the value of the largest number. [6 Marks]
- ✓ 7. Using appropriate examples explain the difference between '*' and '%' operator. [4 Marks]
- ✓ 8. What will be the output of the following program? [2 Marks]

```
void main ( )  
{  
    double x= 23;  
    int r;  
    R = x%5;  
    printf ("R = %d", r);  
}
```

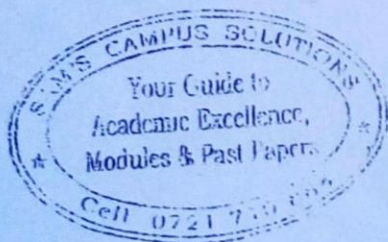


- ✓ 9. What will be the output of the following program? [2 Marks]

```
void main ( )  
{  
    int x=10, y =20;  
    printf ("%d %d", x,y);  
}
```

- ✓ 10. What is the output of the following piece of C code? [2 Marks]

```
int x = 3, y = 5, z;  
z = ++x + ++y;  
printf ("%d", z);
```



[2 Marks]

✓ 11. What is the output of the following piece of C code?

```
int x = 3, y = 5, z;
z = x + 1 + y + 1;
printf("%d", z);
```

[2 Marks]

✓ 12. What is output of following program?

```
int m=1, n=2;
for(j=1; j<=2; j=j+1)
{
    m=m+1;
    n=n+j;
    printf("%d\t%d\n", m, n);
}
```

[6 Marks]

✓ 13. How many times will the loop bodies execute in the following loops?

```
for (int i=1; i<=5; i++)
    for (int j=1; j<=6; j++)
        for (int k=2; k<=5; k++)
            printf("hello\n");
```

```
for (int i=10; i<=12; i++)
    for (int j=1; j<=7; j++)
        printf("hello\n");
```

✓ 14. Which format specifier is used to print the values of double type variable.

[1 Mark]

A) %lf B) %ld C) %lu D) %f

✓ 15. Which of the following is not a keyword of C?

[1 Mark]

A) auto B) register C) int D) function

✓ 16. Which of the following is not a relational operator?

[1 Mark]

A) ! B) != C) <= D) <

✓ 17. What is the valid range of numbers for int type of data?

[1 Mark]

A) 0 to 256 B) -32768 to +32767 C) -65536 to +65536 D) No specific range

✓ 18. The equality operator is represented by:

[1 Mark]

A) == B) EQ C) = D) ==

✓ 19. Operators have precedence. A Precedence determines which operator is

[1 Mark]

A) faster B) takes less memory C) evaluated first D) takes no arguments

⊗ 20. p++ executes faster than ++p because

[1 Mark]

A) p++ uses registers B) p++ is a single instruction C) ++p is faster than ++p D) None of these

EEE 102 14th APRIL 2015

✓ b) Convert the following while loop to an equivalent for loop:

[4 Marks]

```
#include <stdio.h>
```

```
int main()
```

```
{    int x=1;
```

```
    int y;
```

```
    while (x <= 10)
```

```
    {    y=x*x;
```

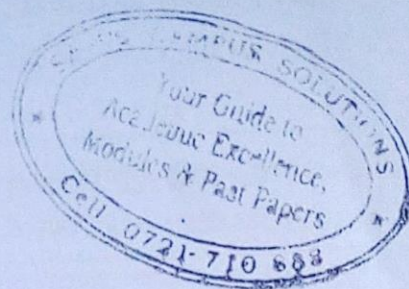
```
        printf("%d %d \n",x,y);
```

```
        x+=3;
```

```
    }
```

```
    return 0;
```

```
}
```



✓ c) Why are variables declared before use?

[2 Marks]

✓ d) How many times will the loop bodies execute in the following loops?

[8 Marks]

i) int x = 4, y = 50;

do

{

 x += 8;

}

while(x < y);

v) for (int i=10; i<12;i++)

 for (int j=3;j<7;j++)

 printf("hello\n");

vi) for (int i=12; i<20;i+=5)

 for (int j=30;j<37;j+=2)

 printf("hello\n");

ii) for (int i=1; i<=5;i+=1)

 for (int j=1;j<=6;j++)

 for (int

 k=2;k<=5;k++)

 printf("hell

o\n");

vii) for (int i=1; i<=3;i+=1)

 for (int j=1;j<=4;j++)

 {

 if (j==3)

 break;

 for (int k=2;k<=4;k++)

 printf("Hello \n");

 }

iii) int y = 0;

for(int x = 5; x < 100; x += 5)

{

 y++;

}

viii) for (int i=1; i<=3;i+=1)

 for (int j=1;j<=4;j++)

 {

 if (j==3)

 continue;

 for (int

 k=2;k<=4;k++)

 printf("Hell

o \n");

 }

iv) int x = 10, y = 1000;

while(x <= y);

{

 x *= 10;

}

Question Three [20 Marks]

- a) Using the monthly income tax table below come up with a flow chart that clearly represents PAYE computation for the system. [8 Marks]

MONTHLY TAXABLE INCOME (INCOME BRACKETS)	TAX ON TAXABLE INCOME	TAX RATE IN EACH SHILLING
For Taxable Income under Kshs. 10,165	Kshs. 1,016 on Taxable Income of Kshs. 10,164	10%
For Taxable Income from Kshs. 10,165 but under Kshs. 19,741	Kshs. 1,016 plus Kshs. 1,436 Tax on Taxable Income of Kshs. 9,576	15%
For Taxable Income from Kshs. 19,741 but under Kshs. 29,317	Kshs. 2,452 plus Kshs. 1,915 Tax on Taxable Income of Kshs. 9,576	20%
For Taxable Income from Kshs. 29,317 but under Kshs. 38,893	Kshs. 4,367 plus Kshs. 2,394 Tax on taxable Income of Kshs. 9,576	25%
For taxable Income from Kshs. 38,893 and above.	Kshs. 6,761 plus Tax calculated at 30% on Taxable Income over Kshs. 38,892	30%

- b) Suppose that you have the following variable declarations:

int x = 10;

int y = 15;

int z = 20;

Determine whether the following expressions are true or false.

[4 Marks]

i. $!(x < 10)$

ii. $x \leq 5 \parallel y > 15$

iii. $(x != 5) \&\& (y == z)$

iv. $x \leq z \&\& (x + y \geq z)$

- c) Giving appropriate examples demonstrate the use of the following operators in C programming: arithmetic operators, logical operators and relational operators. (3 Marks)

- * d) Write a C programming loop that finds the smallest element in an integer array called data containing 100 elements. [3 Marks]

- e) What are data types? Briefly describe any TWO data types in C programming providing details of their format specifiers. [2 Marks]

✓ **Question Four [20 Marks]**

- a) Develop an algorithm for a vending machine controller that sells three products: coke, castle and mineral water at a cost of Kshs. 50.00, Kshs. 30.00 and Kshs. 15.00 respectively. The machine accepts only Kshs. 5.00, 10.00, 20.00, and 40.00 coins as inputs. Change is only available in multiples of Kshs. 5.00. The machine operates as follows: the user inserts sufficient amount of money and presses the product button. The vending machine responds by giving out the selected product and change when due. [7 Marks]

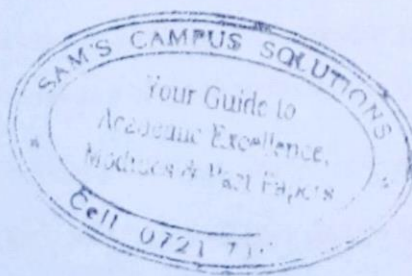
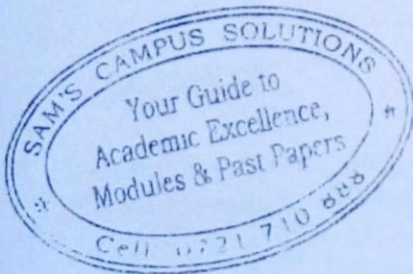
- b) What is the output of the following code fragment? Explain. [3 Marks]

```
int x = 201;
int y = 101;
if (x > 100 && y <= 200)
    printf("%d %d %d", x, y, x + y);
else
    printf("%d %d %d", x, y, 2 * x - y);
```

- c) Briefly describe the following elements of C development environment: *system libraries* and *headers*, *application source*, *compiler*, and *linker*. [4 Marks]

- d) Write a C function that has three inputs which are integers. The function returns true if the first number raised to the power of the second number equals the third number. [3 Marks]

- e) Describe a function in C programming that takes a series of characters in a specific format, converts the characters to data in various forms, and assigns the data to spaces in memory. [3 Marks]



Question Five [20 Marks]

- a) Using the modular program design approach write the algorithm for an ATM machine that strictly allows only the following transactions to be carried out by the user: check account balance, withdraw money, buy airtime and deposit cash. Withdrawals and airtime purchases require that there must be sufficient balance in the account. The system should not allow any ATM transaction of more than KES 20,000. All transactions require PIN authentication and any three consecutive wrong PIN inputs results in retention of the users ATM card [7 Marks]
- b) What is the output from the following C code fragment if the following values are the inputs to scanf: 10, 20, 30, 40, -1. Justify your solutions. [3 Marks]

```
#include <stdio.h>
int main()
{
    int sum, num;
    sum = 0;
    scanf("%d",&num);
    while(num != -1)
    {
        sum += num;
        scanf("%d",&num);
    }
    printf("sum =%d \n",sum);
    return 0;
}
```

- c) Write a C function that has an input of a char value and returns true if the character is lower case or false otherwise. [3 Marks]
- d) Unless specified otherwise, the order of statement execution through a function is linear: one statement after another in sequence. How does C programming allow us to decide whether or not to execute a particular statement or execute a statement over and over, repetitively? [4 Marks]
- e) Explain the main differences between global and local variables in C programming [3 Marks]



KENYATTA UNIVERSITY

School of Engineering

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EEE 102: COMPUTER PROGRAMMING I

CAT 2 [60 Marks]

DATE: 29th March, 2015

TIME: 10h00 - 11h15

1. Provide THREE different methods which can be used to declare an array of size 4 by 3 and type integer in C programming. [6 Marks]



2. Describe the following as used in computer programming:

- ✓ i) Machine language
- ✓ ii) Assembly language
- ✓ iii) High Level Language
- ✓ iv) Source code
- ✓ v) Object code
- * vi) Structured programming
- * vii) Object-oriented programming

[2 Marks]

[2 Marks]

[2 Marks]

[2 Marks]

[2 Marks]

[2 Marks]

[2 Marks]

- ✓ 3. Write the output for the following C program. [6 Marks]

```
#include <stdio.h>
int main()
{
    int a[4][5] = { {1,2,3}, {2}, {1,2,2,2,2}, {32,23} };
    int i,j;

    for (i=0; i<4; i++)
        for (j=0; j<4; j++)
            a[i][j] = a[i][j+1] + i*j + i+j;

    for (i=0; i<4; i++)
    {
        for (j=0; j<5; j++)
            printf(" %d", a[i][j]);
        printf("\n");
    }

    return 0;
}
```

Handwritten output of the C program:

```

      1 2 3 4 5
0      2 3 4 5 6
1      3 4 5 6 7
2      4 5 6 7 8
3      5 6 7 8 9
```

- * 4. What are the advantages of using modular approach in program design? [4 Marks]

5. Briefly describe the following elements of C development environment: system libraries and headers, application source, compiler, and linker. [8 Marks]

- ✓ 6. For each of the following expressions, write the value returned. Assume the following initializations have been made. All problems are valid code and will compile. For boolean values, write either true or false. [7 Marks]
- int w = 1; int x = 2; double y = 1.0; double z = 2.0;

- i) w / x
- ii) 1.0 / z
- iii) (x / w) * (x - w)
- iv) (y / z) / (y / z)
- v) (z + 5.1) >= 6.5 || (x != y)
- vi) w == 2
- vii) x < 9 && (x > 1)

- ✓ 7. Write the output for the following piece of C code. Note that the code is valid, will compile and run. [5 Marks]

```
int x = 3;
int y = 1;

if (x != (2 + y))
    printf("%d,%d\n", x, y);
else
{
    x = 2 + y;
    printf("%d,%d\n", x, y);
    x = 2 + y;
}

for (x = y; x <= (4 + y); x = x + 2)
    printf("%d,%d\n", x, y);
printf("%d,%d\n", x, y);
```

- ✓ 8. Write the output for the following C code. Note that the code is valid, will compile and run. [3 Marks]

```
int a = 16;
int b = 0;

while (a > 1.5 || b < 1.5)
{
    printf("%d,%d\n", a, b);
    a = a / 2;
    b = b + 2;
}

printf("%d,%d\n", a, b);
```

- ★ 9. "C is a compiled language." Explain this statement giving a detailed description of a compiler and the compilation and linking process. [7 Marks]

d) Explain the need for comments and illustrate their position in any C++

[2 Marks]

Question Four [20 Marks]

✓ a) Write the pseudo-code and an algorithm for a C program that sorts the elements of an array in ascending order.

int a[5] = {11, 12, 13, 14, 15};

[7 Marks]

* b) Discuss the generalized methodology involved in the problem solving.

[8 Marks]

study problem
analyze problem
design algorithm
implement
verify result

* c) The Factorial of a number is defined as:

$$\text{factorial}(0) = 1$$

$$\text{factorial}(1) = 1$$

$$\text{factorial of } (n) = n * \text{factorial}(n-1);$$

Write a C++ function to compute the Factorial of any number, n.

[5 Marks]

Question Five [20 Marks]

a) Describe the following operators as used in C/C++ and give examples of each. [8 Marks]

✓ i) Arithmetic Operators +, -, *, /

✓ ii) Relational Operators >, <, >=, <=, ==, !=

✓ iii) Logical Operators &, ||, !

✓ iv) Bitwise Operators &, ||, !

* b) Write a C programming loop that sorts, in ascending, the elements of an integer array called data containing 100 elements.

int data[100];

[6 Marks]

* c) Write C++ code segments consisting of a main function and at least two additional functions to compute the circumference and area of a circle. The program should ask the user to enter a double number corresponding to the radius of the circle. Assuming the lengths of the sides of a square are equivalent to twice the radius of the circle, write additional functions to compute the area and perimeter of the square.

in main().

[6 Marks]

int area, r, s;

s = 2 * r;

A = π * r²;

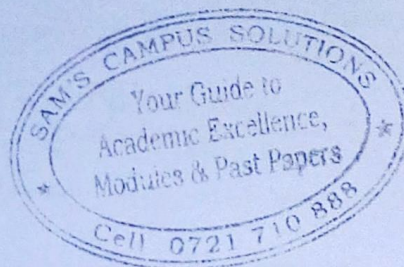
P = 4 * r;



Question Three [20 Marks]

- ✓ a) Write a C program that accepts marks for 10 subjects as input, assigns grades to each of the entered marks and prints out the results in a table form as shown below. [8 Marks]

57: C
45: D
75: A
34: E
63: B



- ✗ b) There are two main forms of repetition, *Fixed Count Loops* and *Variable Count Loops*. Using program code segments draw a distinction between the two. [6 Marks]

- ✓ c) Write an algorithm to determine a student's final grade and indicate whether it is passing or failing. The final grade is calculated as the average of four marks. [4 Marks]

- ✓ d) Determine the number of times that the word "hello" in the following loop will be printed. [2 Marks]

```
for (int i=1; i<=5; i+=1)
    for (int j=1; j<=6; j++)
        for (int k=2; k<=5; k++)
            printf("hello\n");
```

Question Four [20 Marks]

- ✓ a) Determine the validity of the following statement and support your answer using appropriate examples. "Unless specified otherwise, the order of statement execution through a function is linear: one statement after another in sequence." [6 Marks]

- ✓ b) Write an algorithm and draw the flow charts for a program that reads two values, determines the largest value and prints the largest value with an identifying message. [7 Marks]

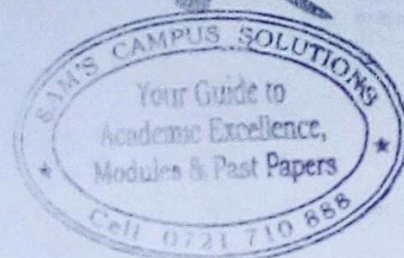
- ✗ c) Write a C program that adds the elements of two 2x2 matrices and returns the solution in as a third matrix. [7 Marks]

Question Five [20 Marks]

- ✓ a) Flowchart is a diagram that shows the step-by-step execution of a control structure. Construct the flow chart for the following nested if statement. [6 Marks]

```
if (road_status == 'S')
    if (temp > 0)
    {
        printf("Wet roads ahead!\n");
    }
    else
    {
        printf("Icy roads ahead!\n");
    }
else
    printf("Drive carefully!\n");
```

- * b) Write a C/C++ program that adds two vectors of the same size. For example: (1,2,3,4,5,6,7,8,9) plus (9,8,7,6,5,4,3,2,1). [8 Marks]
- * c) A positive integer n is said to be prime (or, "a prime") if and only if n is greater than 1 and is divisible only by 1 and n . For example, the integers 17 and 29 are prime, but 1 and 38 are not prime. Write a function named "is_prime" that takes a positive integer argument and returns as its value the integer 1 if the argument is prime and returns the integer 0 otherwise. [6 Marks]



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2016/2017

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(ELECTRICAL AND ELECTRONIC ENGINEERING)

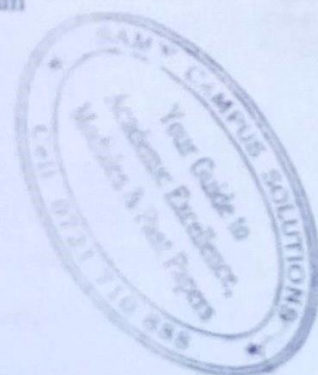
EEE 102: COMPUTER PROGRAMMING I

DATE: MONDAY 5TH JUNE 2017

TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS:

- There are FIVE questions in this exam. Answer Question ONE and any other TWO Questions
- Clearly show your solutions. Neatness and organization in your work is very important
- No electronic device such as calculators and phones may be used in this exam
- Your phone MUST be completely switched off and kept away
- You are not allowed to share any exam materials with other students
- DO NOT place your solutions in a manner that other students can see them



Question One [30 Marks]

- ★ a) Determine the validity of this statement and justify your answer. "C is not strictly a subset of C++, but it is possible to write "Clean C" that conforms to both the C++ and C standards."
[4 Marks]
- ★ b) Explain the role of functions in computer programs and provide the reason why every full C/C++ program begins inside a function called "main".
[4 Marks]
- ✓ c) A data type defines a set of values and a set of operations that can be applied on those values. Briefly describe any TWO C/C++ data types.
[4 Marks]
- ★ d) Explain the concept of "Divide and Conquer" as applied in problem solving using computer algorithms/programs.
[4 Marks]

- 201333
- * e) Distinguish between “a compiler” and “a linker” as used in C/C++ programming. [4 Marks]
 - ✓ f) Describe the term “identifier” as used in C/C++ and provide any FOUR rules for constructing identifiers. [5 Marks]
 - * g) “A *do..while* loop is almost the same as a while loop except that the loop body is guaranteed to execute at least once.” Determine the validity of this statement and justify your answer using appropriate code segment examples. [5 Marks]

Question Two [20 Marks]

- * a) The basic syntax of C can be illustrated by the following example C program. Describe the role and necessity of each of the elements in the program. [8 Marks]

```
#include <stdio.h>
int main()
{
    printf( "HellowWorld\n" );
    getchar();
    return 0;
}
```

- * b) Write a C program that takes in a certain number of integers, say *n*, and computes the total number of occurrences of even numbers in the list that is input. [8 Marks]
- ✓ c) Convert the following *for loop* to an equivalent *while loop*: [4 Marks]

```
#include <stdio.h>
int main()
{
    int y;
    for (int x=1;x<=10;x+=3)
    {
        y=x*x;
        printf("%d %d \n",x,y);
    }
    return 0;
}
```